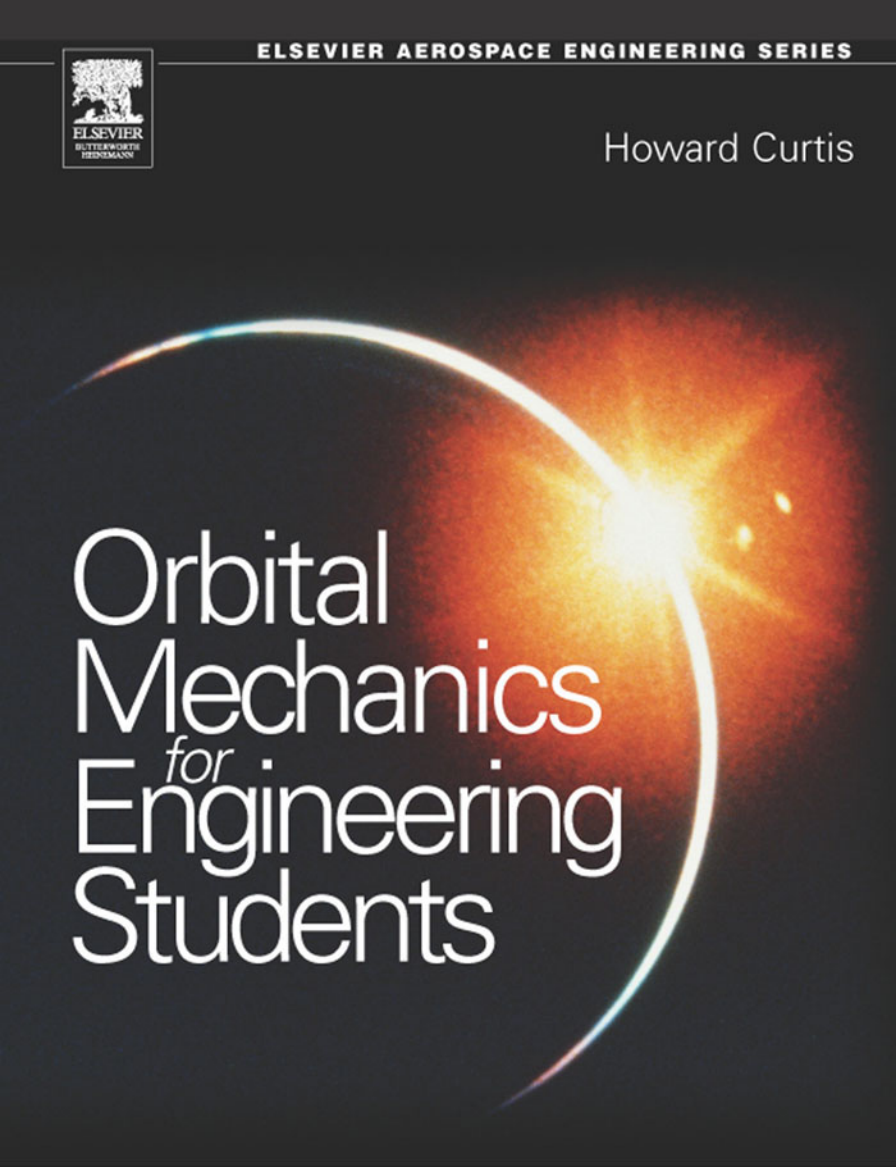




Howard Curtis

The background of the book cover is a dark, deep space scene. A large, bright sun or star is positioned on the right side, casting a strong, warm glow across the scene. A thin, curved line representing the horizon of a planet or moon is visible, separating the dark space from the bright light of the sun. The overall color palette is dominated by the oranges, yellows, and reds of the sun's light, contrasting with the deep blacks and blues of the space.

Orbital Mechanics *for* Engineering Students

ORBITAL MECHANICS FOR ENGINEERING STUDENTS

To my parents, Rondo and Geraldine, and my wife, Connie Dee

ORBITAL MECHANICS FOR ENGINEERING STUDENTS

HOWARD D. CURTIS

*Embry-Riddle Aeronautical University
Daytona Beach, Florida*



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PREFACE

This textbook evolved from a formal set of notes developed over nearly ten years of teaching an introductory course in orbital mechanics for aerospace engineering students. These undergraduate students had no prior formal experience in the subject, but had completed courses in physics, dynamics and mathematics through differential equations and applied linear algebra. That is the background I have presumed for readers of this book.

This is by no means a grand, descriptive survey of the entire subject of astronautics. It is a foundations text, a springboard to advanced study of the subject. I focus on the physical phenomena and analytical procedures required to understand and predict, to first order, the behavior of orbiting spacecraft. I have tried to make the book readable for undergraduates, and in so doing I do not shy away from rigor where it is needed for understanding. Spacecraft operations that take place in earth orbit are considered as are interplanetary missions. The important topic of spacecraft control systems is omitted. However, the material in this book and a course in control theory provide the basis for the study of spacecraft attitude control.

A brief perusal of the Contents shows that there are more than enough topics to cover in a single semester or term. Chapter 1 is a review of vector kinematics in three dimensions and of Newton's laws of motion and gravitation. It also focuses on the issue of relative motion, crucial to the topics of rendezvous and satellite attitude dynamics. Chapter 2 presents the vector-based solution of the classical two-body problem, coming up with a host of practical formulas for orbit and trajectory analysis. The restricted three-body problem is covered in order to introduce the notion of Lagrange points. Chapter 3 derives Kepler's equations, which relate position to time for the different kinds of orbits. The concept of 'universal variables' is introduced. Chapter 4 is devoted to describing orbits in three dimensions and accounting for the major effects of the earth's oblate, non-spherical shape. Chapter 5 is an introduction to preliminary orbit determination, including Gibbs' and Gauss's methods and the solution of Lambert's problem. Auxiliary topics include topocentric coordinate systems, Julian day numbering and sidereal time. Chapter 6 presents the common means of transferring from one orbit to another by impulsive delta- v maneuvers, including Hohmann transfers, phasing orbits and plane changes. Chapter 7 derives and employs the equations of relative motion required to understand and design two-impulse rendezvous maneuvers. Chapter 8 explores the basics of interplanetary mission analysis. Chapter 9 presents those elements of rigid-body dynamics required to characterize the attitude of an orbiting satellite. Chapter 10 describes the methods of controlling, changing and stabilizing the attitude of spacecraft by means of thrusters, gyros and other devices. Finally, Chapter 11 is a brief introduction to the characteristics and design of multi-stage launch vehicles.

Chapters 1 through 4 form the core of a first orbital mechanics course. The time devoted to Chapter 1 depends on the background of the student. It might be surveyed

briefly and used thereafter simply as a reference. What follows Chapter 4 depends on the objectives of the course.

Chapters 5 through 8 carry on with the subject of orbital mechanics. Chapter 6 on orbital maneuvers should be included in any case. Coverage of Chapters 5, 7 and 8 is optional. However, if all of Chapter 8 on interplanetary missions is to form a part of the course, then the solution of Lambert's problem (Section 5.3) must be studied beforehand.

Chapters 9 and 10 must be covered if the course objectives include an introduction to satellite dynamics. In that case Chapters 5, 7 and 8 would probably not be studied in depth.

Chapter 11 is optional if the engineering curriculum requires a separate course in propulsion, including rocket dynamics.

To understand the material and to solve problems requires using a lot of undergraduate mathematics. Mathematics, of course, is the language of engineering. Students must not forget that Sir Isaac Newton had to invent calculus so he could solve orbital mechanics problems precisely. Newton (1642–1727) was an English physicist and mathematician, whose 1687 publication *Mathematical Principles of Natural Philosophy* ('the *Principia*') is one of the most influential scientific works of all time. It must be noted that the German mathematician Gottfried Wilhelm von Leibniz (1646–1716) is credited with inventing infinitesimal calculus independently of Newton in the 1670s.

In addition to honing their math skills, students are urged to take advantage of computers (which, incidentally, use the binary numeral system developed by Leibniz). There are many commercially available mathematics software packages for personal computers. Wherever possible they should be used to relieve the burden of repetitive and tedious calculations. Computer programming skills can and should be put to good use in the study of orbital mechanics. Elementary MATLAB® programs (M-files) appear at the end of this book to illustrate how some of the procedures developed in the text can be implemented in software. All of the scripts were developed using MATLAB version 5.0 and were successfully tested using version 6.5 (release 13). Information about MATLAB, which is a registered trademark of The MathWorks, Inc., may be obtained from:

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The text contains many detailed explanations and worked-out examples. Their purpose is not to overwhelm but to elucidate. It is always assumed that the material is being seen for the first time and, wherever possible, solution details are provided so as to leave little to the reader's imagination. There are some exceptions to this objective, deemed necessary to maintain the focus and control the size of the book. For example, in Chapter 6, the notion of specific impulse is laid on the table as a means of rating rocket motor performance and to show precisely how delta-v is related to propellant expenditure. In Chapter 10 Routh–Hurwitz stability criteria are used without proof to

show quantitatively that a particular satellite configuration is, indeed, stable. Specific impulse is covered in more detail in Chapter 11, and the stability of linear systems is treated in depth in books on control theory. See, for example, Nise (2003) and Ogata (2001).

Supplementary material appears in the appendices at the end of the book. Appendix A lists physical data for use throughout the text. Appendix B is a ‘road map’ to guide the reader through Chapters 1, 2 and 3. Appendix C shows how to set up the n -body equations of motion and program them in MATLAB. Appendix D lists the MATLAB implementations of algorithms presented in several of the chapters. Appendix E shows that the gravitational field of a spherically symmetric body is the same as if the mass were concentrated at its center.

The field of astronautics is rich and vast. References cited throughout this text are listed at the end of the book. Also listed are other books on the subject that might be of interest to those seeking additional insights.

I wish to thank colleagues who provided helpful criticism and advice during the development of this book. Yechiel Crispin and Charles Eastlake were sources for ideas about what should appear in the summary chapter on rocket dynamics. Habib Eslami, Lakshmanan Narayanaswami, Mahmut Reyhanoglu and Axel Rohde all used the evolving manuscript as either a text or a reference in their space mechanics courses. Based on their classroom experiences, they gave me valuable feedback in the form of corrections, recommendations and much-needed encouragement. Tony Hagar voluntarily and thoroughly reviewed the entire manuscript and made a number of suggestions, nearly all of which were incorporated into the final version of the text.

I am indebted to those who reviewed the manuscript for the publisher for their many suggestions on how the book could be improved and what additional topics might be included.

Finally, let me acknowledge how especially grateful I am to the students who, throughout the evolution of the book, reported they found it to be a helpful and understandable introduction to space mechanics.

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SUPPLEMENTS TO THE TEXT

For the student:

- Copies of the MATLAB programs (M-files) that appear in Appendix D can be downloaded from the companion website accompanying this book. To access these please visit <http://books.elsevier.com/companions> and follow the instructions on screen.

For the instructor:

- A full *Instructor's Solutions Manual* is available for adopting tutors, which provides complete worked-out solutions to the problems set at the end of each chapter. To access these please visit <http://books.elsevier.com/manuals> and follow the instructions on screen.

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DYNAMICS OF POINT MASSES

CHAPTER OUTLINE

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1.3	MASS, FORCE AND NEWTON'S LAW OF GRAVITATION	7
1.4	NEWTON'S LAW OF MOTION	10
1.5	TIME DERIVATIVES OF MOVING VECTORS	15
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	PROBLEMS	29

1.1 INTRODUCTION

This chapter serves as a self-contained reference on the kinematics and dynamics of point masses as well as some basic vector operations. The notation and concepts summarized here will be used in the following chapters. Those familiar with the vector-based dynamics of particles can simply page through the chapter and then refer back to it later as necessary. Those who need a bit more in the way of review will find the chapter contains all of the material they need in order to follow the development of orbital mechanics topics in the upcoming chapters.

We begin with the problem of describing the curvilinear motion of particles in three dimensions. The concepts of force and mass are considered next, along with Newton's inverse-square law of gravitation. This is followed by a presentation

of Newton's second law of motion ('force equals mass times acceleration') and the important concept of angular momentum.

As a prelude to describing motion relative to moving frames of reference, we develop formulas for calculating the time derivatives of moving vectors. These are applied to the computation of relative velocity and acceleration. Example problems illustrate the use of these results as does a detailed consideration of how the earth's rotation and curvature influence our measurements of velocity and acceleration. This brings in the curious concept of Coriolis force. Embedded in exercises at the end of the chapter is practice in verifying several fundamental vector identities that will be employed frequently throughout the book.

1.2 KINEMATICS

To track the motion of a particle P through Euclidean space we need a frame of reference, consisting of a clock and a cartesian coordinate system. The clock keeps track of time t and the xyz axes of the cartesian coordinate system are used to locate the spatial position of the particle. In non-relativistic mechanics, a single 'universal' clock serves for all possible cartesian coordinate systems. So when we refer to a frame of reference we need think only of the mutually orthogonal axes themselves.

The unit of time used throughout this book is the second (s). The unit of length is the meter (m), but the kilometer (km) will be the length unit of choice when large distances and velocities are involved. Conversion factors between kilometers, miles and nautical miles are listed in Table A.3.

Given a frame of reference, the position of the particle P at a time t is defined by the position vector $\mathbf{r}(t)$ extending from the origin O of the frame out to P itself, as illustrated in Figure 1.1. (Vectors will always be indicated by boldface type.) The

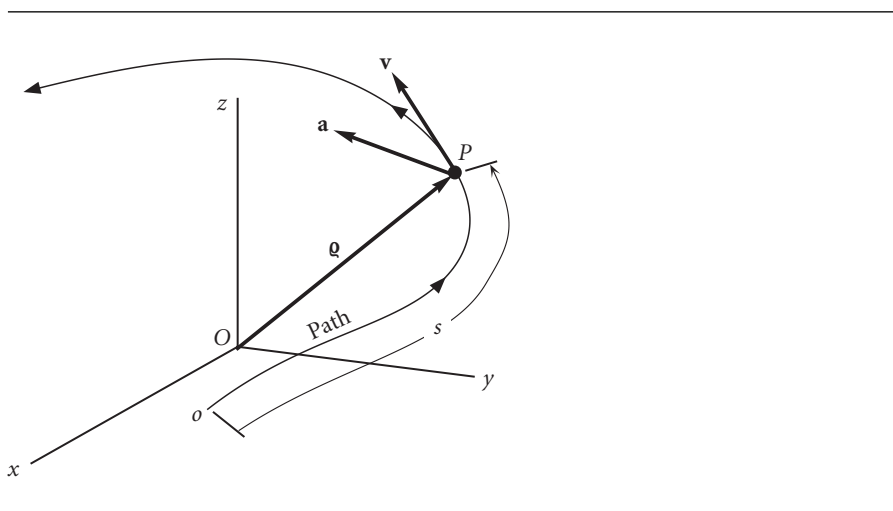


Figure 1.1 Position, velocity and acceleration vectors.

components of $\mathbf{r}(t)$ are just the x , y and z coordinates,

$$\mathbf{r}(t) = x(t)\hat{\mathbf{i}} + y(t)\hat{\mathbf{j}} + z(t)\hat{\mathbf{k}}$$

$\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ are the unit vectors which point in the positive direction of the x , y and z axes, respectively. Any vector written with the overhead hat (e.g., $\hat{\mathbf{a}}$) is to be considered a vector of unit dimensionless magnitude.

The distance of P from the origin is the magnitude or length of $\mathbf{r}$, denoted $\|\mathbf{r}\|$ or just r ,

$$\|\mathbf{r}\| = r = \sqrt{x^2 + y^2 + z^2}$$

The magnitude of $\mathbf{r}$, or any vector $\mathbf{A}$ for that matter, can also be computed by means of the dot product operation,

$$r = \sqrt{\mathbf{r} \cdot \mathbf{r}} \quad \|\mathbf{A}\| = \sqrt{\mathbf{A} \cdot \mathbf{A}}$$

The velocity $\mathbf{v}$ and acceleration $\mathbf{a}$ of the particle are the first and second time derivatives of the position vector,

$$\mathbf{v}(t) = \frac{dx(t)}{dt}\hat{\mathbf{i}} + \frac{dy(t)}{dt}\hat{\mathbf{j}} + \frac{dz(t)}{dt}\hat{\mathbf{k}} = v_x(t)\hat{\mathbf{i}} + v_y(t)\hat{\mathbf{j}} + v_z(t)\hat{\mathbf{k}}$$

$$\mathbf{a}(t) = \frac{dv_x(t)}{dt}\hat{\mathbf{i}} + \frac{dv_y(t)}{dt}\hat{\mathbf{j}} + \frac{dv_z(t)}{dt}\hat{\mathbf{k}} = a_x(t)\hat{\mathbf{i}} + a_y(t)\hat{\mathbf{j}} + a_z(t)\hat{\mathbf{k}}$$

It is convenient to represent the time derivative by means of an overhead dot. In this shorthand notation, if $()$ is any quantity, then

$$(\dot{}) \equiv \frac{d()}{dt} \quad (\ddot{}) \equiv \frac{d^2()}{dt^2} \quad (\dddot{}) \equiv \frac{d^3()}{dt^3}, \text{ etc.}$$

Thus, for example,

$$\begin{aligned} \mathbf{v} &= \dot{\mathbf{r}} \\ \mathbf{a} &= \dot{\mathbf{v}} = \ddot{\mathbf{r}} \\ v_x &= \dot{x} & v_y &= \dot{y} & v_z &= \dot{z} \\ a_x &= \dot{v}_x = \ddot{x} & a_y &= \dot{v}_y = \ddot{y} & a_z &= \dot{v}_z = \ddot{z} \end{aligned}$$

The locus of points that a particle occupies as it moves through space is called its path or trajectory. If the path is a straight line, then the motion is rectilinear. Otherwise, the path is curved, and the motion is called curvilinear. The velocity vector $\mathbf{v}$ is tangent to the path. If $\hat{\mathbf{u}}_t$ is the unit vector tangent to the trajectory, then

$$\mathbf{v} = v\hat{\mathbf{u}}_t$$

where v , the speed, is the magnitude of the velocity $\mathbf{v}$. The distance ds that P travels along its path in the time interval dt is obtained from the speed by

$$ds = v dt$$

In other words,

$$v = \dot{s}$$

The distance s , measured along the path from some starting point, is what the odometers in our automobiles record. Of course, $\dot{s}$, our speed along the road, is indicated by the dial of the speedometer.

Note carefully that $v \neq \dot{r}$, i.e., the magnitude of the derivative of $\mathbf{r}$ does not equal the derivative of the magnitude of $\mathbf{r}$.

EXAMPLE 1.1

The position vector in meters is given as a function of time in seconds as

$$\mathbf{r} = (8t^2 + 7t + 6)\hat{\mathbf{i}} + (5t^3 + 4)\hat{\mathbf{j}} + (0.3t^4 + 2t^2 + 1)\hat{\mathbf{k}} \text{ (m)} \quad (\text{a})$$

At $t = 10$ seconds, calculate v (the magnitude of the derivative of $\mathbf{r}$) and $\dot{r}$ (the derivative of the magnitude of $\mathbf{r}$).

The velocity $\mathbf{v}$ is found by differentiating the given position vector with respect to time,

$$\mathbf{v} = \frac{d\mathbf{r}}{dt} = (16t + 7)\hat{\mathbf{i}} + 15t^2\hat{\mathbf{j}} + (1.2t^3 + 4t)\hat{\mathbf{k}}$$

The magnitude of this vector is the square root of the sum of the squares of its components,

$$\|\mathbf{v}\| = (1.44t^6 + 234.6t^4 + 272t^2 + 224t + 49)^{\frac{1}{2}}$$

Evaluating this at $t = 10$ s, we get

$$\underline{v = 1953.3 \text{ m/s}}$$

Calculating the magnitude of $\mathbf{r}$ in (a), leads to

$$\|\mathbf{r}\| = (0.09t^8 + 26.2t^6 + 68.6t^4 + 152t^3 + 149t^2 + 84t + 53)^{\frac{1}{2}}$$

Differentiating this expression with respect to time,

$$\dot{r} = \frac{dr}{dt} = \frac{0.36t^7 + 78.6t^5 + 137.2t^3 + 228t^2 + 149t + 42}{(0.09t^8 + 26.2t^6 + 68.6t^4 + 152t^3 + 149t^2 + 84t + 53)^{\frac{1}{2}}}$$

Substituting $t = 10$ s, yields

$$\underline{\dot{r} = 1935.5 \text{ m/s}}$$

If $\mathbf{v}$ is given, then we can find the components of the unit tangent $\hat{\mathbf{u}}_t$ in the cartesian coordinate frame of reference

$$\hat{\mathbf{u}}_t = \frac{\mathbf{v}}{\|\mathbf{v}\|} = \frac{v_x}{v}\hat{\mathbf{i}} + \frac{v_y}{v}\hat{\mathbf{j}} + \frac{v_z}{v}\hat{\mathbf{k}} \quad \left(v = \sqrt{v_x^2 + v_y^2 + v_z^2}\right)$$

The acceleration may be written,

$$\mathbf{a} = a_t \hat{\mathbf{u}}_t + a_n \hat{\mathbf{u}}_n$$

where a_t and a_n are the tangential and normal components of acceleration, given by

$$a_t = \dot{v} (= \ddot{s}) \quad a_n = \frac{v^2}{\varrho} \quad (1.1)$$

ϱ is the radius of curvature, which is the distance from the particle P to the center of curvature of the path at that point. The unit principal normal $\hat{\mathbf{u}}_n$ is perpendicular to $\hat{\mathbf{u}}_t$ and points towards the center of curvature C , as shown in Figure 1.2. Therefore, the position of C relative to P , denoted $\mathbf{r}_{C/P}$, is

$$\mathbf{r}_{C/P} = \varrho \hat{\mathbf{u}}_n$$

The orthogonal unit vectors $\hat{\mathbf{u}}_t$ and $\hat{\mathbf{u}}_n$ form a plane called the osculating plane. The unit normal to the osculating plane is $\hat{\mathbf{u}}_b$, the binormal, and it is obtained from $\hat{\mathbf{u}}_t$ and $\hat{\mathbf{u}}_n$ by taking their cross product,

$$\hat{\mathbf{u}}_b = \hat{\mathbf{u}}_t \times \hat{\mathbf{u}}_n$$

The center of curvature lies in the osculating plane. When the particle P moves an incremental distance ds the radial from the center of curvature to the path sweeps out a small angle $d\phi$, measured in the osculating plane. The relationship between this angle and ds is

$$ds = \varrho d\phi$$

so that $\dot{s} = \varrho \dot{\phi}$, or

$$\dot{\phi} = \frac{v}{\varrho} \quad (1.2)$$

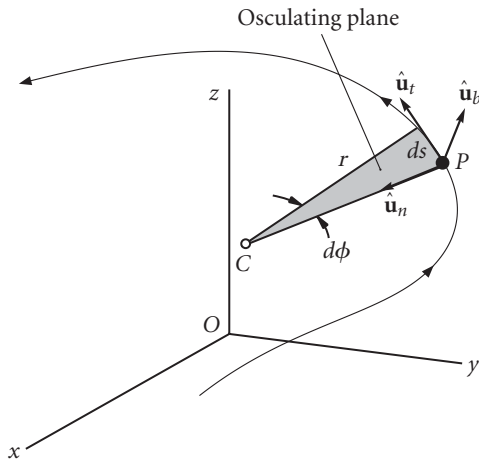


Figure 1.2 Orthogonal triad of unit vectors associated with the moving point P .

EXAMPLE
1.2

Relative to a cartesian coordinate system, the position, velocity and acceleration of a particle relative at a given instant are

$$\mathbf{r} = 250\hat{\mathbf{i}} + 630\hat{\mathbf{j}} + 430\hat{\mathbf{k}} \text{ (m)}$$

$$\mathbf{v} = 90\hat{\mathbf{i}} + 125\hat{\mathbf{j}} + 170\hat{\mathbf{k}} \text{ (m/s)}$$

$$\mathbf{a} = 16\hat{\mathbf{i}} + 125\hat{\mathbf{j}} + 30\hat{\mathbf{k}} \text{ (m/s}^2\text{)}$$

Find the coordinates of the center of curvature at that instant.

First, we calculate the speed v ,

$$v = \|\mathbf{v}\| = \sqrt{90^2 + 125^2 + 170^2} = 229.4 \text{ m/s}$$

The unit tangent is, therefore,

$$\hat{\mathbf{u}}_t = \frac{\mathbf{v}}{v} = \frac{90\hat{\mathbf{i}} + 125\hat{\mathbf{j}} + 170\hat{\mathbf{k}}}{229.4} = 0.3923\hat{\mathbf{i}} + 0.5449\hat{\mathbf{j}} + 0.7411\hat{\mathbf{k}}$$

We project the acceleration vector onto the direction of the tangent to get its tangential component a_t ,

$$a_t = \mathbf{a} \cdot \hat{\mathbf{u}}_t = (16\hat{\mathbf{i}} + 125\hat{\mathbf{j}} + 30\hat{\mathbf{k}}) \cdot (0.3923\hat{\mathbf{i}} + 0.5449\hat{\mathbf{j}} + 0.7411\hat{\mathbf{k}}) = 96.62 \text{ m/s}^2$$

The magnitude of $\mathbf{a}$ is

$$a = \sqrt{16^2 + 125^2 + 30^2} = 129.5 \text{ m/s}^2$$

Since $\mathbf{a} = a_t \hat{\mathbf{u}}_t + a_n \hat{\mathbf{u}}_n$ and $\hat{\mathbf{u}}_t$ and $\hat{\mathbf{u}}_n$ are perpendicular to each other, it follows that $a^2 = a_t^2 + a_n^2$, which means

$$a_n = \sqrt{a^2 - a_t^2} = \sqrt{129.5^2 - 96.62^2} = 86.29 \text{ m/s}^2$$

Hence,

$$\begin{aligned} \hat{\mathbf{u}}_n &= \frac{1}{a_n}(\mathbf{a} - a_t \hat{\mathbf{u}}_t) \\ &= \frac{1}{86.29}[(16\hat{\mathbf{i}} + 125\hat{\mathbf{j}} + 30\hat{\mathbf{k}}) - 96.62(0.3923\hat{\mathbf{i}} + 0.5449\hat{\mathbf{j}} + 0.7411\hat{\mathbf{k}})] \\ &= -0.2539\hat{\mathbf{i}} + 0.8385\hat{\mathbf{j}} - 0.4821\hat{\mathbf{k}} \end{aligned}$$

The equation $a_n = v^2/\varrho$ can now be solved for ϱ to yield

$$\varrho = \frac{v^2}{a_n} = \frac{229.4^2}{86.29} = 609.9 \text{ m}$$

Let $\mathbf{r}_C$ be the position vector of the center of curvature C . Then

$$\begin{aligned}\mathbf{r}_C &= \mathbf{r} + \mathbf{r}_{C/P} \\ &= \mathbf{r} + \rho \hat{\mathbf{u}}_n = 250\hat{\mathbf{i}} + 630\hat{\mathbf{j}} + 430\hat{\mathbf{k}} + 609.9(-0.2539\hat{\mathbf{i}} + 0.8385\hat{\mathbf{j}} - 0.4821\hat{\mathbf{k}}) \\ &= 95.16\hat{\mathbf{i}} + 1141\hat{\mathbf{j}} + 136.0\hat{\mathbf{k}} \text{ (m)}\end{aligned}$$

That is, the coordinates of C are

$$\underline{x = 95.16 \text{ m}} \quad \underline{y = 1141 \text{ m}} \quad \underline{z = 136.0 \text{ m}}$$

1.3 MASS, FORCE AND NEWTON'S LAW OF GRAVITATION

Mass, like length and time, is a primitive physical concept: it cannot be defined in terms of any other physical concept. Mass is simply the quantity of matter. More practically, mass is a measure of the inertia of a body. Inertia is an object's resistance to changing its state of motion. The larger its inertia (the greater its mass), the more difficult it is to set a body into motion or bring it to rest. The unit of mass is the kilogram (kg).

Force is the action of one physical body on another, either through direct contact or through a distance. Gravity is an example of force acting through a distance, as are magnetism and the force between charged particles. The gravitational force between two masses m_1 and m_2 having a distance r between their centers is

$$F_g = G \frac{m_1 m_2}{r^2} \quad (1.3)$$

This is Newton's law of gravity, in which G , the universal gravitational constant, has the value $6.6742 \times 10^{-11} \text{ m}^3/\text{kg} \cdot \text{s}^2$. Due to the inverse-square dependence on distance, the force of gravity rapidly diminishes with the amount of separation between the two masses. In any case, the force of gravity is minuscule unless at least one of the masses is extremely big.

The force of a large mass (such as the earth) on a mass many orders of magnitude smaller (such as a person) is called weight, W . If the mass of the large object is M and that of the relatively tiny one is m , then the weight of the small body is

$$W = G \frac{Mm}{r^2} = m \left(\frac{GM}{r^2} \right)$$

or

$$W = mg \quad (1.4)$$

where

$$g = \frac{GM}{r^2} \quad (1.5)$$

g has units of acceleration (m/s^2) and is called the acceleration of gravity. If planetary gravity is the only force acting on a body, then the body is said to be in free fall. The force of gravity draws a freely falling object towards the center of attraction (e.g., center of the earth) with an acceleration g . Under ordinary conditions, we sense our own weight by feeling contact forces acting on us in opposition to the force of gravity. In free fall there are, by definition, no contact forces, so there can be no sense of weight. Even though the weight is not zero, a person in free fall experiences weightlessness, or the absence of gravity.

Let us evaluate Equation 1.5 at the surface of the earth, whose radius according to Table A.1 is 6378 km. Letting g_0 represent the standard sea-level value of g , we get

$$g_0 = \frac{GM}{R_E^2} \quad (1.6)$$

In SI units,

$$g_0 = 9.807 \text{ m/s}^2 \quad (1.7)$$

Substituting Equation 1.6 into Equation 1.5 and letting z represent the distance above the earth's surface, so that $r = R_E + z$, we obtain

$$g = g_0 \frac{R_E^2}{(R_E + z)^2} = \frac{g_0}{(1 + z/R_E)^2} \quad (1.8)$$

Commercial airliners cruise at altitudes on the order of 10 kilometers (six miles). At that height, Equation 1.8 reveals that g (and hence weight) is only three-tenths of a percent less than its sea-level value. Thus, under ordinary conditions, we ignore the variation of g with altitude. A plot of Equation 1.8 out to a height of 1000 km (the upper limit of low-earth orbit operations) is shown in Figure 1.3. The variation of g over that range is significant. Even so, at space station altitude (300 km), weight is only about 10 percent less than it is on the earth's surface. The astronauts experience weightlessness, but they clearly are not weightless.

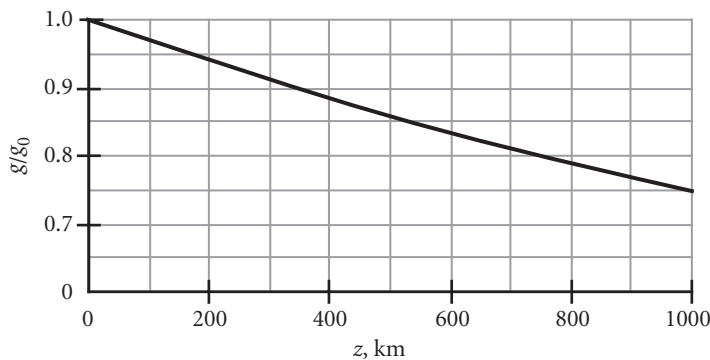
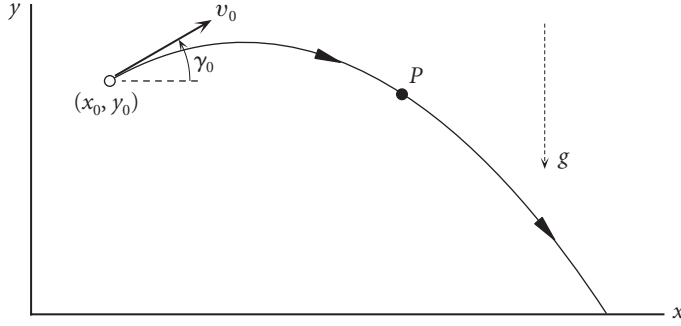


Figure 1.3 Variation of the acceleration of gravity with altitude.

EXAMPLE
1.3

Show that in the absence of an atmosphere, the shape of a low altitude ballistic trajectory is a parabola. Assume the acceleration of gravity g is constant and neglect the earth's curvature.

**Figure 1.4**

Flight of a low altitude projectile in free fall (no atmosphere).

Figure 1.4 shows a projectile launched at $t = 0$ with a speed v_0 at a flight path angle γ_0 from the point with coordinates (x_0, y_0) . Since the projectile is in free fall after launch, its only acceleration is that of gravity in the negative y -direction:

$$\ddot{x} = 0$$

$$\ddot{y} = -g$$

Integrating with respect to time and applying the initial conditions leads to

$$x = x_0 + (v_0 \cos \gamma_0)t \quad (a)$$

$$y = y_0 + (v_0 \sin \gamma_0)t - \frac{1}{2}gt^2 \quad (b)$$

Solving (a) for t and substituting the result into (b) yields

$$y = y_0 + (x - x_0) \tan \gamma_0 - \frac{1}{2} \frac{g}{v_0^2 \cos^2 \gamma_0} (x - x_0)^2 \quad (c)$$

This is the equation of a second-degree curve, a parabola, as sketched in Figure 1.4.

EXAMPLE
1.4

An airplane flies a parabolic trajectory like that in Figure 1.4 so that the passengers will experience free fall (weightlessness). What is the required variation of the flight path angle γ with speed v ? Ignore the curvature of the earth.

Figure 1.5 reveals that for a 'flat' earth, $d\gamma = -d\phi$, i.e.,

$$\dot{\gamma} = -\dot{\phi}$$

(Example 1.4
continued)

It follows from Equation 1.2 that

$$\varrho \dot{\gamma} = -v \quad (1.9)$$

The normal acceleration a_n is just the component of the gravitational acceleration g in the direction of the unit principal normal to the curve (from P towards C). From Figure 1.5, then,

$$a_n = g \cos \gamma \quad (a)$$

Substituting Equation 1.1 into (a) and solving for the radius of curvature yields

$$\varrho = \frac{v^2}{g \cos \gamma} \quad (b)$$

Combining Equations 1.9 and (b), we find the time rate of change of the flight path angle,

$$\dot{\gamma} = -\frac{g \cos \gamma}{v}$$

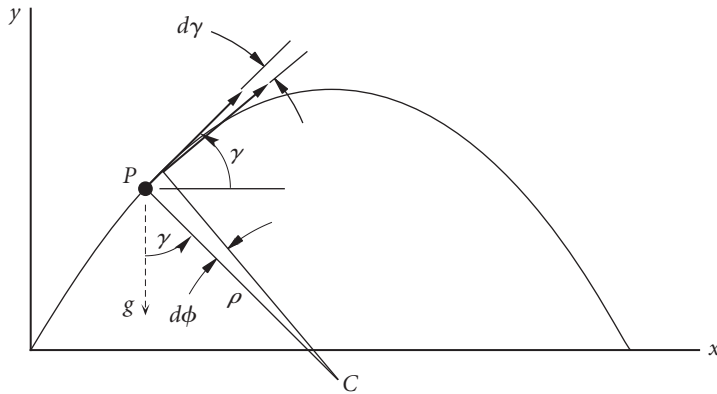


Figure 1.5 Relationship between $d\gamma$ and $d\phi$ for a 'flat' earth.

1.4 NEWTON'S LAW OF MOTION

Force is not a primitive concept like mass because it is intimately connected with the concepts of motion and inertia. In fact, the only way to alter the motion of a body is to exert a force on it. The degree to which the motion is altered is a measure of the force. This is quantified by Newton's second law of motion. If the resultant or net force on a body of mass m is $\mathbf{F}_{\text{net}}$, then

$$\mathbf{F}_{\text{net}} = m\mathbf{a} \quad (1.10)$$

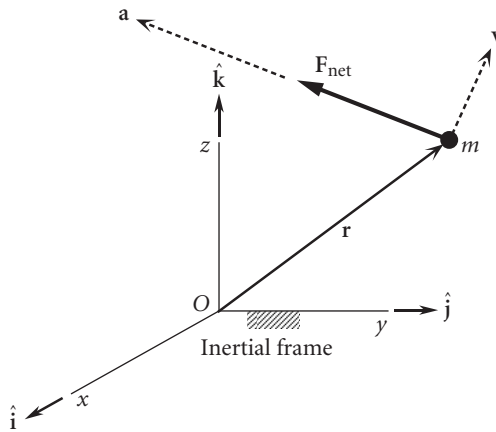


Figure 1.6 The absolute acceleration of a particle is in the direction of the net force.

In this equation, $\mathbf{a}$ is the absolute acceleration of the center of mass. The absolute acceleration is measured in a frame of reference which itself has neither translational nor rotational acceleration relative to the fixed stars. Such a reference is called an absolute or inertial frame of reference.

Force, then, is related to the primitive concepts of mass, length and time by Newton's second law. The unit of force, appropriately, is the Newton, which is the force required to impart an acceleration of 1 m/s^2 to a mass of 1 kg . A mass of one kilogram therefore weighs 9.81 Newtons at the earth's surface. The kilogram is not a unit of force.

Confusion can arise when mass is expressed in units of force, as frequently occurs in US engineering practice. In common parlance either the pound or the ton (2000 pounds) is more likely to be used to express the mass. The pound of mass is officially defined precisely in terms of the kilogram as shown in Table A.3. Since one pound of mass weighs one pound of force where the standard sea-level acceleration of gravity ($g_0 = 9.80665 \text{ m/s}^2$) exists, we can use Newton's second law to relate the pound of force to the Newton:

$$\begin{aligned} 1 \text{ lb (force)} &= 0.4536 \text{ kg} \times 9.807 \text{ m/s}^2 \\ &= 4.448 \text{ N} \end{aligned}$$

The slug is the quantity of matter accelerated at one foot per second² by a force of one pound. We can again use Newton's second law to relate the slug to the kilogram. Noting the relationship between feet and meters in Table A.3, we find

$$\begin{aligned} 1 \text{ slug} &= \frac{1 \text{ lb}}{1 \text{ ft/s}^2} = \frac{4.448 \text{ N}}{0.3048 \text{ m/s}^2} = 14.59 \frac{\text{kg} \cdot \text{m/s}^2}{\text{m/s}^2} \\ &= 14.59 \text{ kg} \end{aligned}$$

EXAMPLE
1.5

On a NASA mission the space shuttle Atlantis orbiter was reported to weigh 239 255 lb just prior to lift-off. On orbit 18 at an altitude of about 350 km, the orbiter's weight was reported to be 236 900 lb. (a) What was the mass, in kilograms, of Atlantis on the launch pad and in orbit? (b) If no mass were lost between launch and orbit 18, what would have been the weight of Atlantis in pounds?

- (a) The given data illustrates the common use of weight in pounds as a measure of mass. The 'weights' given are actually the mass in pounds of mass. Therefore, prior to launch

$$m_{\text{launch pad}} = 239\,255 \text{ lb (mass)} \times \frac{0.4536 \text{ kg}}{1 \text{ lb (mass)}} = \underline{108\,500 \text{ kg}}$$

In orbit,

$$m_{\text{orbit 18}} = 236\,900 \text{ lb (mass)} \times \frac{0.4536 \text{ kg}}{1 \text{ lb (mass)}} = \underline{107\,500 \text{ kg}}$$

The decrease in mass is the propellant expended by the orbital maneuvering and reaction control rockets on the orbiter.

- (b) Since the space shuttle launch pad at Kennedy Space Center is essentially at sea level, the launch-pad weight of Atlantis in lb (force) is numerically equal to its mass in lb (mass). With no change in mass, the force of gravity at 350 km would be, according to Equation 1.8,

$$W = 239\,255 \text{ lb (force)} \times \left(\frac{1}{1 + \frac{350}{6378}} \right)^2 = \underline{215\,000 \text{ lb (force)}}$$

The integral of a force $\mathbf{F}$ over a time interval is called the impulse I of the force,

$$\mathbf{I} = \int_{t_1}^{t_2} \mathbf{F} dt \quad (1.11)$$

From Equation 1.10 it is apparent that if the mass is constant, then

$$\mathbf{I}_{\text{net}} = \int_{t_1}^{t_2} m \frac{d\mathbf{v}}{dt} dt = m\mathbf{v}_2 - m\mathbf{v}_1 \quad (1.12)$$

That is, the net impulse on a body yields a change $m\Delta\mathbf{v}$ in its linear momentum, so that

$$\Delta\mathbf{v} = \frac{\mathbf{I}_{\text{net}}}{m} \quad (1.13)$$

If $\mathbf{F}_{\text{net}}$ is constant, then $\mathbf{I}_{\text{net}} = \mathbf{F}_{\text{net}}\Delta t$, in which case Equation 1.13 becomes

$$\Delta\mathbf{v} = \frac{\mathbf{F}_{\text{net}}}{m} \Delta t \quad (\text{if } \mathbf{F}_{\text{net}} \text{ is constant}) \quad (1.14)$$

Let us conclude this section by introducing the concept of angular momentum. The moment of the net force about O in Figure 1.6 is

$$\mathbf{M}_{O_{\text{net}}} = \mathbf{r} \times \mathbf{F}_{\text{net}}$$

Substituting Equation 1.10 yields

$$\mathbf{M}_{O_{\text{net}}} = \mathbf{r} \times m\mathbf{a} = \mathbf{r} \times m \frac{d\mathbf{v}}{dt} \quad (1.15)$$

But, keeping in mind that the mass is constant,

$$\mathbf{r} \times m \frac{d\mathbf{v}}{dt} = \frac{d}{dt}(\mathbf{r} \times m\mathbf{v}) - \left(\frac{d\mathbf{r}}{dt} \times m\mathbf{v} \right) = \frac{d}{dt}(\mathbf{r} \times m\mathbf{v}) - (\mathbf{v} \times m\mathbf{v})$$

Since $\mathbf{v} \times m\mathbf{v} = m(\mathbf{v} \times \mathbf{v}) = \mathbf{0}$, it follows that Equation 1.15 can be written

$$\mathbf{M}_{O_{\text{net}}} = \frac{d\mathbf{H}_O}{dt} \quad (1.16)$$

where $\mathbf{H}_O$ is the angular momentum about O ,

$$\mathbf{H}_O = \mathbf{r} \times m\mathbf{v} \quad (1.17)$$

Thus, just as the net force on a particle changes its linear momentum $m\mathbf{v}$, the moment of that force about a fixed point changes the moment of its linear momentum about that point. Integrating Equation 1.16 with respect to time yields

$$\int_{t_1}^{t_2} \mathbf{M}_{O_{\text{net}}} dt = \mathbf{H}_{O_2} - \mathbf{H}_{O_1} \quad (1.18)$$

The integral on the left is the net angular impulse. This angular impulse–momentum equation is the rotational analog of the linear impulse–momentum relation given above in Equation 1.12.

EXAMPLE 1.6

A particle of mass m is attached to point O by an inextensible string of length l . Initially the string is slack when m is moving to the left with a speed v_0 in the position shown. Calculate the speed of m just after the string becomes taut. Also, compute the

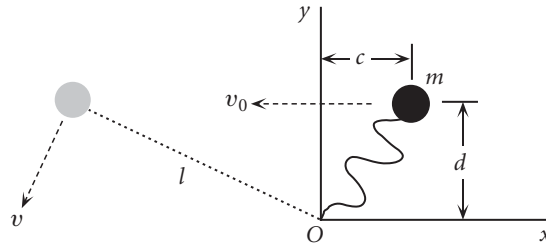


Figure 1.7

Particle attached to O by an inextensible string.

(Example 1.6
continued)

average force in the string over the small time interval Δt required to change the direction of the particle's motion.

Initially, the position and velocity of the particle are

$$\mathbf{r}_1 = c\hat{\mathbf{i}} + d\hat{\mathbf{j}} \quad \mathbf{v}_1 = -v_0\hat{\mathbf{i}}$$

The angular momentum is

$$\mathbf{H}_1 = \mathbf{r}_1 \times m\mathbf{v}_1 = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ c & d & 0 \\ -mv_0 & 0 & 0 \end{vmatrix} = mv_0\hat{\mathbf{k}} \quad (\text{a})$$

Just after the string becomes taut

$$\mathbf{r}_2 = -\sqrt{l^2 - d^2}\hat{\mathbf{i}} + d\hat{\mathbf{j}} \quad \mathbf{v}_2 = v_x\hat{\mathbf{i}} + v_y\hat{\mathbf{j}} \quad (\text{b})$$

and the angular momentum is

$$\mathbf{H}_2 = \mathbf{r}_2 \times m\mathbf{v}_2 = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ -\sqrt{l^2 - d^2} & d & 0 \\ v_x & v_y & 0 \end{vmatrix} = (-mv_xd - mv_y\sqrt{l^2 - d^2})\hat{\mathbf{k}} \quad (\text{c})$$

Initially the force exerted on m by the slack string is zero. When the string becomes taut, the force exerted on m passes through O . Therefore, the moment of the net force on m about O remains zero. According to Equation 1.18,

$$\mathbf{H}_2 = \mathbf{H}_1$$

Substituting (a) and (c) yields

$$v_xd + \sqrt{l^2 - d^2}v_y = -v_0d \quad (\text{d})$$

The string is inextensible, so the component of the velocity of m along the string must be zero:

$$\mathbf{v}_2 \cdot \mathbf{r}_2 = 0$$

Substituting $\mathbf{v}_2$ and $\mathbf{r}_2$ from (b) and solving for v_y we get

$$v_y = v_x\sqrt{\frac{l^2}{d^2} - 1} \quad (\text{e})$$

Solving (d) and (e) for v_x and v_y leads to

$$v_x = -\frac{d^2}{l^2}v_0 \quad v_y = -\sqrt{1 - \frac{d^2}{l^2}}\frac{d}{l}v_0 \quad (\text{f})$$

Thus, the speed, $v = \sqrt{v_x^2 + v_y^2}$, after the string becomes taut is

$$\underline{v = \frac{d}{l}v_0}$$

From Equation 1.12, the impulse on m during the time it takes the string to become taut is

$$\begin{aligned}\mathbf{I} &= m(\mathbf{v}_2 - \mathbf{v}_1) = m \left[\left(-\frac{d^2}{l^2} v_0 \hat{\mathbf{i}} - \sqrt{1 - \frac{d^2}{l^2}} \frac{d}{l} v_0 \hat{\mathbf{j}} \right) - (-v_0 \hat{\mathbf{i}}) \right] \\ &= \left(1 - \frac{d^2}{l^2} \right) m v_0 \hat{\mathbf{i}} - \sqrt{1 - \frac{d^2}{l^2}} \frac{d}{l} m v_0 \hat{\mathbf{j}}\end{aligned}$$

The magnitude of this impulse, which is directed along the string, is

$$I = \sqrt{1 - \frac{d^2}{l^2}} m v_0$$

Hence, the average force in the string during the small time interval Δt required to change the direction of the velocity vector turns out to be

$$F_{\text{avg}} = \frac{I}{\Delta t} = \sqrt{1 - \frac{d^2}{l^2}} \frac{m v_0}{\Delta t}$$

1.5 TIME DERIVATIVES OF MOVING VECTORS

Figure 1.8(a) shows a vector $\mathbf{A}$ inscribed in a rigid body B that is in motion relative to an inertial frame of reference (a rigid, cartesian coordinate system which is fixed relative to the fixed stars). The magnitude of $\mathbf{A}$ is fixed. The body B is shown at two times, separated by the differential time interval dt . At time $t + dt$ the orientation of

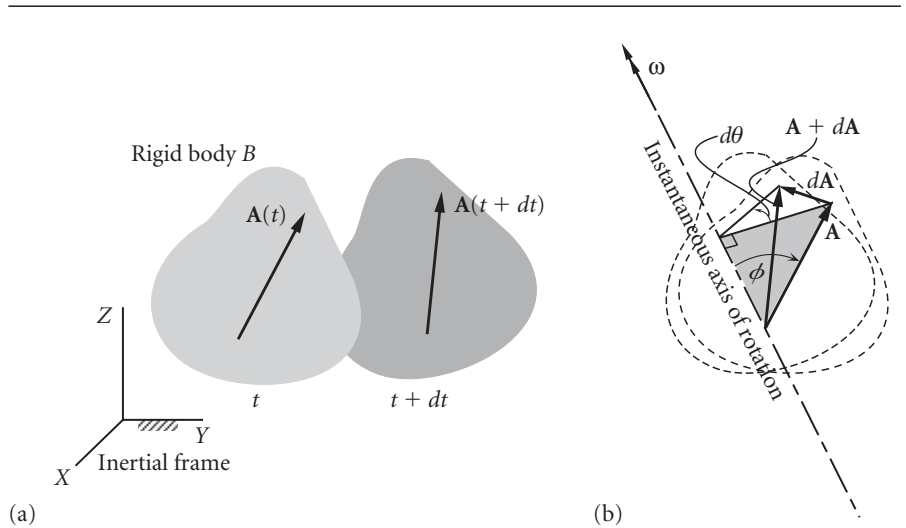


Figure 1.8 Displacement of a rigid body.

vector $\mathbf{A}$ differs slightly from that at time t , but its magnitude is the same. According to one of the many theorems of the prolific eighteenth century Swiss mathematician Leonhard Euler (1707–1783), there is a unique axis of rotation about which B and, therefore, $\mathbf{A}$ rotates during the differential time interval. If we shift the two vectors $\mathbf{A}(t)$ and $\mathbf{A}(t + dt)$ to the same point on the axis of rotation, so that they are tail-to-tail as shown in Figure 1.8(b), we can assess the difference $d\mathbf{A}$ between them caused by the infinitesimal rotation. Remember that shifting a vector to a parallel line does not change the vector. The rotation of the body B is measured in the plane perpendicular to the instantaneous axis of rotation. The amount of rotation is the angle $d\theta$ through which a line element normal to the rotation axis turns in the time interval dt . In Figure 1.8(b) that line element is the component of $\mathbf{A}$ normal to the axis of rotation. We can express the difference $d\mathbf{A}$ between $\mathbf{A}(t)$ and $\mathbf{A}(t + dt)$ as

$$d\mathbf{A} = \overbrace{[(\|\mathbf{A}\| \cdot \sin \phi) d\theta]}^{\text{magnitude of } d\mathbf{A}} \hat{\mathbf{n}} \quad (1.19)$$

where $\hat{\mathbf{n}}$ is the unit normal to the plane defined by $\mathbf{A}$ and the axis of rotation, and it points in the direction of the rotation. The angle ϕ is the inclination of $\mathbf{A}$ to the rotation axis. By definition,

$$d\theta = \|\boldsymbol{\omega}\| dt \quad (1.20)$$

where $\boldsymbol{\omega}$ is the angular velocity vector, which points along the instantaneous axis of rotation and its direction is given by the right-hand rule. That is, wrapping the right hand around the axis of rotation, with the fingers pointing in the direction of $d\theta$, results in the thumb's defining the direction of $\boldsymbol{\omega}$. This is evident in Figure 1.8(b). It should be pointed out that the time derivative of $\boldsymbol{\omega}$ is the angular acceleration, usually given the symbol $\boldsymbol{\alpha}$. Thus,

$$\boldsymbol{\alpha} = \frac{d\boldsymbol{\omega}}{dt} \quad (1.21)$$

Substituting Equation 1.20 into Equation 1.19, we get

$$d\mathbf{A} = \|\mathbf{A}\| \cdot \sin \phi \|\boldsymbol{\omega}\| dt \cdot \hat{\mathbf{n}} = (\|\boldsymbol{\omega}\| \cdot \|\mathbf{A}\| \cdot \sin \phi) \hat{\mathbf{n}} dt \quad (1.22)$$

By definition of the cross product, $\boldsymbol{\omega} \times \mathbf{A}$ is the product of the magnitude of $\boldsymbol{\omega}$, the magnitude of $\mathbf{A}$, the sine of the angle between $\boldsymbol{\omega}$ and $\mathbf{A}$ and the unit vector normal to the plane of $\boldsymbol{\omega}$ and $\mathbf{A}$, in the rotation direction. That is,

$$\boldsymbol{\omega} \times \mathbf{A} = \|\boldsymbol{\omega}\| \cdot \|\mathbf{A}\| \cdot \sin \phi \cdot \hat{\mathbf{n}} \quad (1.23)$$

Substituting Equation 1.23 into Equation 1.22 yields

$$d\mathbf{A} = \boldsymbol{\omega} \times \mathbf{A} dt$$

Dividing through by dt , we finally obtain

$$\frac{d\mathbf{A}}{dt} = \boldsymbol{\omega} \times \mathbf{A} \quad (1.24)$$

Equation 1.24 is a formula we can use to compute the time derivative of any vector of constant magnitude.

EXAMPLE
1.7

Calculate the second time derivative of a vector $\mathbf{A}$ of constant magnitude, expressing the result in terms of $\boldsymbol{\omega}$ and its derivatives and $\mathbf{A}$.

Differentiating Equation 1.24 with respect to time, we get

$$\frac{d^2\mathbf{A}}{dt^2} = \frac{d}{dt} \frac{d\mathbf{A}}{dt} = \frac{d}{dt}(\boldsymbol{\omega} \times \mathbf{A}) = \frac{d\boldsymbol{\omega}}{dt} \times \mathbf{A} + \boldsymbol{\omega} \times \frac{d\mathbf{A}}{dt}$$

Using Equations 1.21 and 1.24, this can be written

$$\frac{d^2\mathbf{A}}{dt^2} = \boldsymbol{\alpha} \times \mathbf{A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{A}) \quad (1.25)$$

EXAMPLE
1.8

Calculate the third derivative of a vector $\mathbf{A}$ of constant magnitude, expressing the result in terms of $\boldsymbol{\omega}$ and its derivatives and $\mathbf{A}$.

$$\begin{aligned} \frac{d^3\mathbf{A}}{dt^3} &= \frac{d}{dt} \frac{d^2\mathbf{A}}{dt^2} = \frac{d}{dt} [\boldsymbol{\alpha} \times \mathbf{A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{A})] \\ &= \frac{d}{dt}(\boldsymbol{\alpha} \times \mathbf{A}) + \frac{d}{dt}[\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{A})] \\ &= \left(\frac{d\boldsymbol{\alpha}}{dt} \times \mathbf{A} + \boldsymbol{\alpha} \times \frac{d\mathbf{A}}{dt} \right) + \left[\frac{d\boldsymbol{\omega}}{dt} \times (\boldsymbol{\omega} \times \mathbf{A}) + \boldsymbol{\omega} \times \frac{d}{dt}(\boldsymbol{\omega} \times \mathbf{A}) \right] \\ &= \left[\frac{d\boldsymbol{\alpha}}{dt} \times \mathbf{A} + \boldsymbol{\alpha} \times (\boldsymbol{\omega} \times \mathbf{A}) \right] + \left[\boldsymbol{\alpha} \times (\boldsymbol{\omega} \times \mathbf{A}) + \boldsymbol{\omega} \times \left(\frac{d\boldsymbol{\omega}}{dt} \times \mathbf{A} + \boldsymbol{\omega} \times \frac{d\mathbf{A}}{dt} \right) \right] \\ &= \left[\frac{d\boldsymbol{\alpha}}{dt} \times \mathbf{A} + \boldsymbol{\alpha} \times (\boldsymbol{\omega} \times \mathbf{A}) \right] + \{ \boldsymbol{\alpha} \times (\boldsymbol{\omega} \times \mathbf{A}) + \boldsymbol{\omega} \times [\boldsymbol{\alpha} \times \mathbf{A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{A})] \} \\ &= \frac{d\boldsymbol{\alpha}}{dt} \times \mathbf{A} + \boldsymbol{\alpha} \times (\boldsymbol{\omega} \times \mathbf{A}) + \boldsymbol{\alpha} \times (\boldsymbol{\omega} \times \mathbf{A}) + \boldsymbol{\omega} \times (\boldsymbol{\alpha} \times \mathbf{A}) + \boldsymbol{\omega} \times [\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{A})] \\ &= \frac{d\boldsymbol{\alpha}}{dt} \times \mathbf{A} + 2\boldsymbol{\alpha} \times (\boldsymbol{\omega} \times \mathbf{A}) + \boldsymbol{\omega} \times (\boldsymbol{\alpha} \times \mathbf{A}) + \boldsymbol{\omega} \times [\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{A})] \\ \frac{d^3\mathbf{A}}{dt^3} &= \frac{d\boldsymbol{\alpha}}{dt} \times \mathbf{A} + 2\boldsymbol{\alpha} \times (\boldsymbol{\omega} \times \mathbf{A}) + \boldsymbol{\omega} \times [\boldsymbol{\alpha} \times \mathbf{A} + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{A})] \end{aligned}$$

Let XYZ be a rigid inertial frame of reference and xyz a rigid moving frame of reference, as shown in Figure 1.9. The moving frame can be moving (translating and rotating) freely of its own accord, or it can be imagined to be attached to a physical object, such as a car, an airplane or a spacecraft. Kinematic quantities measured relative to the fixed inertial frame will be called absolute (e.g., absolute acceleration), and those measured relative to the moving system will be called relative (e.g., relative acceleration). The unit vectors along the inertial XYZ system are $\hat{\mathbf{I}}, \hat{\mathbf{J}}$ and $\hat{\mathbf{K}}$, whereas those of the moving xyz system are $\hat{\mathbf{i}}, \hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$. The motion of the moving frame is arbitrary, and its absolute angular velocity is $\boldsymbol{\Omega}$. If, however, the moving frame is rigidly attached to an object, so that it not only translates but rotates with it, then the

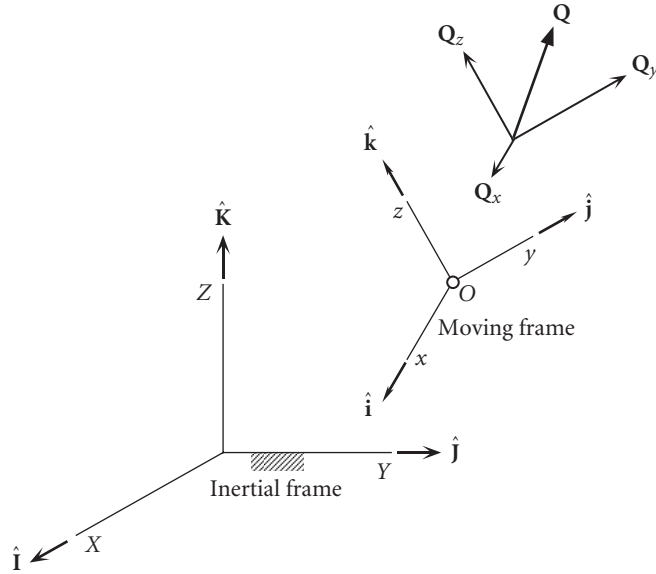


Figure 1.9 Fixed (inertial) and moving rigid frames of reference.

frame is called a body frame and the axes are referred to as body axes. A body frame clearly has the same angular velocity as the body to which it is bound.

Let $\mathbf{Q}$ be any time-dependent vector. Resolved into components along the inertial frame of reference, it is expressed analytically as

$$\mathbf{Q} = Q_X \hat{\mathbf{i}} + Q_Y \hat{\mathbf{j}} + Q_Z \hat{\mathbf{k}}$$

where Q_X , Q_Y and Q_Z are functions of time. Since $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ are fixed, the time derivative of $\mathbf{Q}$ is simply given by

$$\frac{d\mathbf{Q}}{dt} = \frac{dQ_X}{dt} \hat{\mathbf{i}} + \frac{dQ_Y}{dt} \hat{\mathbf{j}} + \frac{dQ_Z}{dt} \hat{\mathbf{k}}$$

dQ_X/dt , dQ_Y/dt and dQ_Z/dt are the components of the absolute time derivative of $\mathbf{Q}$.

$\mathbf{Q}$ may also be resolved into components along the moving xyz frame, so that, at any instant,

$$\mathbf{Q} = Q_x \hat{\mathbf{i}} + Q_y \hat{\mathbf{j}} + Q_z \hat{\mathbf{k}} \quad (1.26)$$

Using this expression to calculate the time derivative of $\mathbf{Q}$ yields

$$\frac{d\mathbf{Q}}{dt} = \frac{dQ_x}{dt} \hat{\mathbf{i}} + \frac{dQ_y}{dt} \hat{\mathbf{j}} + \frac{dQ_z}{dt} \hat{\mathbf{k}} + Q_x \frac{d\hat{\mathbf{i}}}{dt} + Q_y \frac{d\hat{\mathbf{j}}}{dt} + Q_z \frac{d\hat{\mathbf{k}}}{dt} \quad (1.27)$$

The unit vectors $\hat{\mathbf{i}}$, $\hat{\mathbf{j}}$ and $\hat{\mathbf{k}}$ are not fixed in space, but are continuously changing direction; therefore, their time derivatives are not zero. They obviously have a constant